

COP2034C Module 2 Programming Assignment

Instructions

In this assignment, you will expand the Palmer Penguins program you created in Module 1 by adding variables, numeric data, arithmetic expressions, floating-point calculations, and use of the Python math module.

We will continue working with the three species represented in the Palmer Penguins dataset:

- Chinstrap
- Gentoo
- Adelie

The Palmer Penguins data is included as a .csv file in your GitHub starter repository. We are not yet going to read the file with Python. For now, we will use known values from the dataset to perform several basic calculations.

As you work, continue to develop your program incrementally: add a small amount of code, run it, verify that it works correctly, and then continue.

Development Environment

Complete this assignment in **Google Colab** (begin with the Palmer Penguins program you completed for Module 1). After your program executes correctly:

1. Download your notebook as a Python (.py) file.
2. Upload the .py file to your forked GitHub repository.
3. Take a screenshot showing the successful execution of your program in Google Colab.
4. Save the screenshot as a .png or .jpg file.
5. Upload the screenshot to your GitHub repository.

Do not submit the .ipynb notebook.

1. Start with Your Module 1 Program

Open the Palmer Penguins notebook you created for Module 1.

Rename the notebook for Module 2. A suggested name is:

PalmerPenguinsM2.ipynb

The Python file you eventually download should have a meaningful name such as:

PalmerPenguinsM2.py

2. Update the ID Header

Update the four-line ID header at the top of your program.

```
# PalmerPenguinsM2.py
# Your Name
# Submission Date
# Calculates and displays Palmer Penguin statistics.
```

Leave a blank line after the header.

3. Retain the Existing Constants

Keep the constants from Module 1 that represent the three species and the number of species in the dataset.

```
SP_CHINSTRAP = "Chinstrap"
SP_GENTOO = "Gentoo"
SP_ADELIE = "Adelie"
```

```
TOTAL_SPECIES = 3
```

Remember that Python does not enforce constants. By convention, we use **ALL UPPERCASE names** for values that should not change while the program executes.

4. Add Constants for the Number of Penguins

The Palmer Penguins dataset contains observations for 342 penguins.

Add three new constants representing the number of observations for each species:

```
NUM_CHINSTRAP = 68
NUM_GENTOO = 123
NUM_ADELIE = 151
```

5. Calculate the Total Number of Penguins

Create a variable named **total_penguins**.

Calculate its value by adding the three species-count constants:

```
total_penguins = NUM_CHINSTRAP + NUM_GENTOO + NUM_ADELIE
```

Do **not** simply assign 342 to the variable. The purpose is to practice using variables and arithmetic expressions.

Notice the naming conventions used in the program:

- Constants use UPPERCASE_NAMES.
- Regular variables use lowercase names with underscores.

6. Calculate the Average Number of Penguins per Species

Create a variable named `average_penguins`.

Calculate the average number of observations per species by dividing `total_penguins` by

`TOTAL_SPECIES`:

```
average_penguins = total_penguins / TOTAL_SPECIES
```

The `/` operator performs division and produces a floating-point result.

For example:

```
10 / 4
```

produces:

```
2.5
```

7. Calculate the Percentage of Each Species

Create the following variables:

```
chinstrap_percent
```

```
gentoo_percent
```

```
adelie_percent
```

The general calculation for a percentage is:

```
part / whole * 100
```

For example, calculate the Chinstrap percentage using:

```
chinstrap_percent = NUM_CHINSTRAP / total_penguins * 100
```

Create similar expressions for Gentoo and Adelie.

These calculations produce floating-point values. **Do not round or specially format the percentages in this assignment.** Display the floating-point values as Python calculates them. We will learn additional techniques for formatting output in a later module.

8. Use the math Module

Python includes modules that provide additional functions for use in our programs.

Add the following statement near the beginning of your program, after the ID header:

```
import math
```

Use the `math.ceil()` function to determine the smallest whole number greater than or equal to the average number of penguins per species.

Store the result in a variable named `average_penguins_rounded_up`:

```
average_penguins_rounded_up = math.ceil(average_penguins)
```

This gives you practice importing a Python module and using one of its functions.

9. Display the Results

Retain the introductory output from Module 1:

Introducing the Palmer Penguins:

```
Chinstrap!  
Gentoo!  
and last but not least...
```

```
Adelie!
```

Then display:

- the number of species,
- the total number of penguins,
- the average number of penguins per species,
- the average rounded up to a whole number,
- the count and percentage for each species.

Continue using the same output techniques introduced in Module 1:

- `print()`
- string concatenation using `+`
- `str()` to convert numeric values to strings

For example:

```
print("There are a total of " +  
      str(total_penguins) +  
      " penguins in the dataset.")
```

To display a species count and percentage:

```
print(SP_CHINSTRAP + ": " +  
      str(NUM_CHINSTRAP) + " (" +  
      str(chinstrap_percent) + "%)")
```

Notice how the long `print()` statement is split across multiple lines to keep each line at 80 characters or fewer.

Expected Output

Your program should produce output similar to the following:

Introducing the Palmer Penguins:

```
Chinstrap!  
Gentoo!  
and last but not least...
```

Adelie!

There are a total of 3 penguin species in this dataset.

There are a total of 342 penguins in the dataset.

The average number of penguins per species is 114.0.

The average rounded up is 114 penguins.

Chinstrap: 68 (19.883040935672515%)

Gentoo: 123 (35.96491228070175%)

Adelie: 151 (44.15204678362573%)

The percentage values are intentionally displayed without special formatting. You will learn additional techniques for controlling the appearance of floating-point output in a later module.

Data Science Connection

Open the Palmer Penguins .csv file provided in your GitHub repository and briefly examine the data.

Notice:

- the number of rows, or **observations**,
- the columns, or **variables**,
- the different types of values stored in the columns,
- the three penguin species represented in the data.

Examining a dataset before beginning an analysis is an important part of **Exploratory Data Analysis (EDA)**.

In later modules, we will use Python to work with the dataset directly. For now, the species counts allow us to begin performing simple descriptive calculations with the data.

Readability

In addition to whether your program runs correctly, you will be graded on code readability.

Keep your program readable by:

- using meaningful variable names,
- using consistent whitespace around operators,
- separating logical sections with blank lines,
- keeping lines reasonably short,
- adding comments where appropriate,
- following the Python naming conventions demonstrated in this assignment.

Use comments to identify logical sections of your program, such as:

```
# constants representing the species and count
```

```
# calculate dataset statistics  
# output the species names with introductory text
```

Screenshot

Run the completed program in Google Colab.

Take a screenshot showing the program's **output** and successful execution.

The screenshot does not need to include all of your source code because the .py file will be reviewed separately.

Save the screenshot as a .png or .jpg file.

A suggested filename is:

PalmerPenguinsM2.png

Submission

Upload the following two files to your forked GitHub repository:

- PalmerPenguinsM2.py
- Screenshot of the program executing successfully (.png or .jpg)

Submit the URL of your forked GitHub repository to the Canvas assignment.

Do not submit the Google Colab .ipynb notebook.